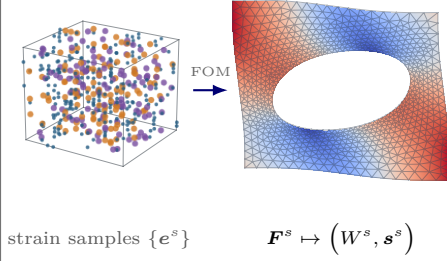
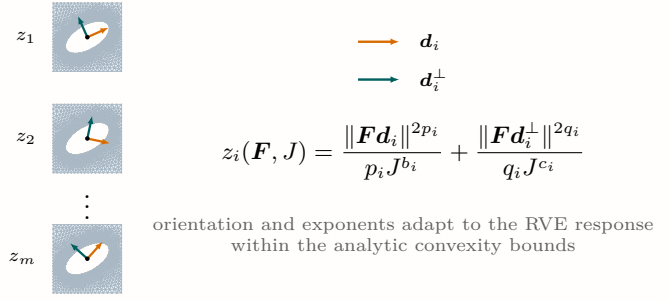


1 Sampling and RVE data

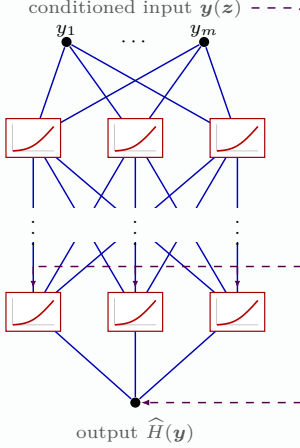


2 Learnable paired directional features

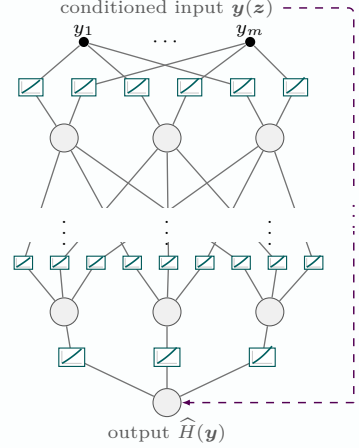


3 Monotone convex core

ICNN
fixed softplus activations on nodes



ICKAN
learned convex, nondecreasing functions on edges



$H(z) = \hat{H}(y(z))$
learned contribution

4 Complete stored energy

$$W(F) = \underbrace{H(z) - H(z^0)}_{W_{\text{learn}}} - \underbrace{r(J-1)}_{W_{\text{corr}}} + \underbrace{\alpha(J-1 - \log J) + \frac{\beta}{2}(J-1)^2}_{W_{\text{vol}}} + \underbrace{\varepsilon \left(\frac{\text{tr } C}{2} - 1 - \log J \right)}_{W_{\text{dist}}}$$

reference state: $W(I) = 0, \quad s(0) = 0$

5 Constitutive output

$$\underbrace{s = \nabla_e W}_{\text{stress}}$$

$$\underbrace{D = \nabla_e^2 W}_{\text{tangent}}$$

Criteria satisfied by construction

(C1) potential structure

(C2) objectivity

(C3) stress symmetry and anisotropy

(C4) reference normalization

(C5) 2D polyconvexity

(C6) growth